

Ex.no:6c	DATA VISUALIZATION USING POWER BI (E-COMMERCE ANALYTICS)

Aim:

To clean, preprocess, and transform the raw E-Commerce dataset, load the cleaned data into Power BI, and build an interactive visual dashboard to analyze purchases, quantities, unit prices, customer behavior, and product category patterns.

Algorithm:

1. Import Dataset: Load E_Commerce.csv into the environment.
2. Deduplication: Identify and delete duplicate rows across all fields.
3. Remove Blank Rows: Identify and remove completely blank records from the dataset.
4. Handle Missing Values: Identify missing values in each column and replace or remove them appropriately to maintain data consistency.
5. Handle Data Type Errors: Convert numerical attributes such as age, unit_price, quantity, purchase_frequency, and cancellations_count into appropriate numerical data types.
6. Clean Text Data: Remove unnecessary spaces and standardize categorical values such as country, product_name, category, gender, and subscription_status.
7. Create Purchase Calculation: Calculate the total purchase value using the unit price and quantity.

Total Purchase Value = unit_price × quantity

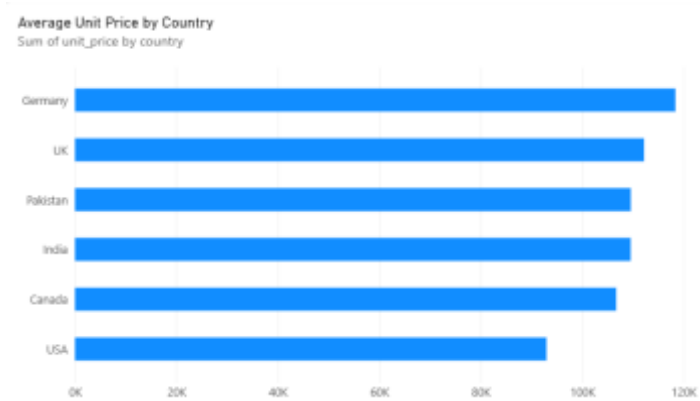
8. Load Clean Data: Load the cleaned E-Commerce dataset into Power BI Desktop using Get Data → Text/CSV and select Transform Data if further cleaning is required.
9. Create Core Metric: Create a measure in Power BI to calculate the total purchase value.

Total Purchase Value = SUMX(E_Commerce, E_Commerce[unit_price] * E_Commerce[quantity])

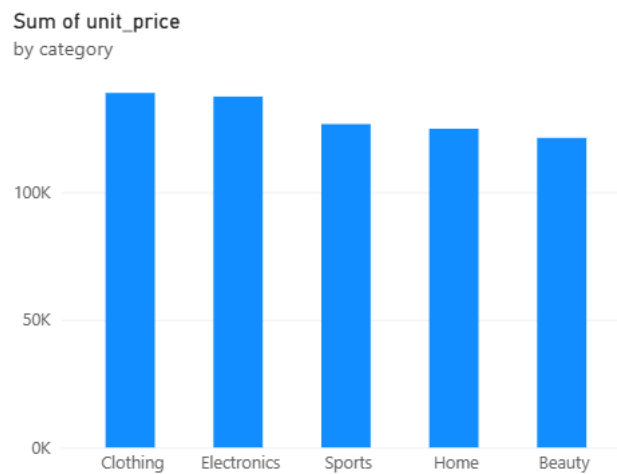
10. Create Dashboard: Build interactive visualizations such as KPI cards, bar charts, pie charts, line charts, and slicers to analyze country-wise, category-wise, and customer-wise purchase patterns.

Output:

1.BAR CHART

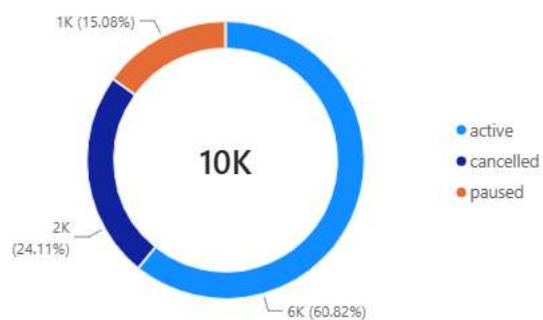


2.Column Chart



3.Donut Chart

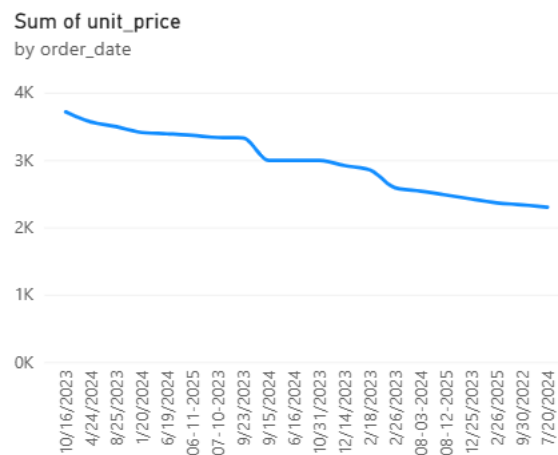
Sum of quantity
by subscription_status



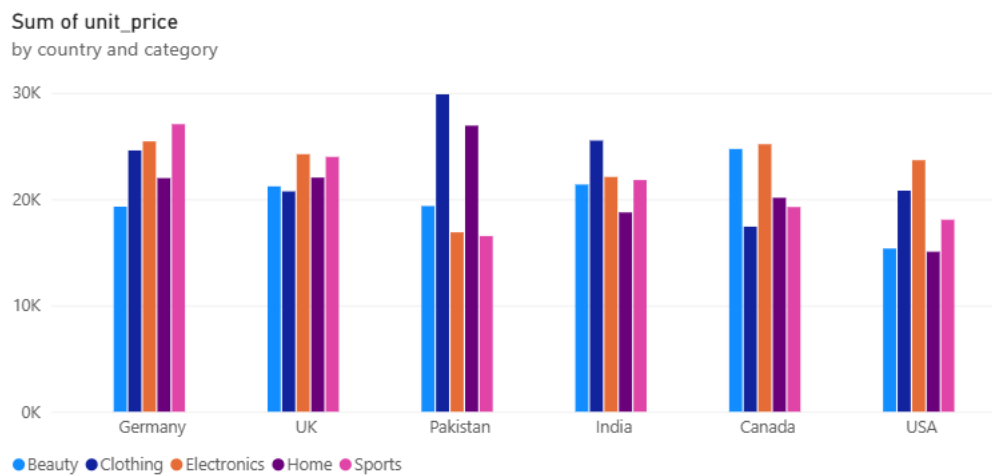
4.KPI CARD



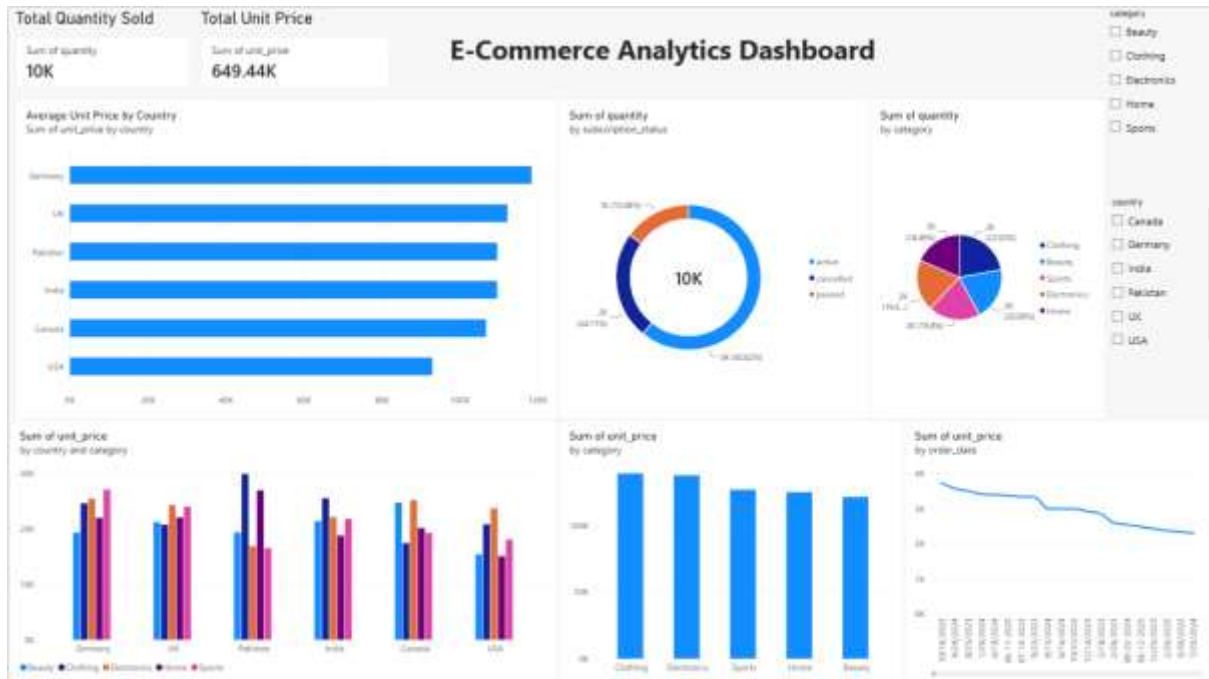
5.Line Chart



6.Clustered Column Chart



7.DASHBOARD



Result: